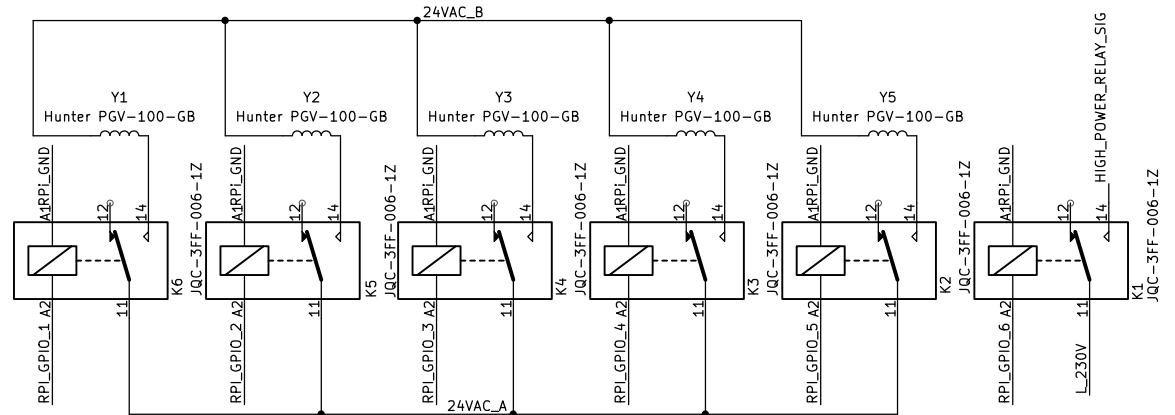
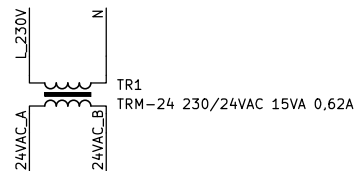


Note to pump power connection:

When using pump also check carefully AC-7b of your high power relay. That value should be ideally slightly bigger than your pump wattage. For example EZCT-20-20-230 has AC-7b at 7A, so it can be used to pumps with power of 1.2 kW

Obviously you have to ground your pump if it has metal parts exposed.

It is good practice to also add water flow sensor, so it can prevent pump from damaging itself when valve for any reason doesn't open and pump is working.



Note:

Depending on your relay board, you might connect JD-VCC to 5V instead 3.3V. Optoisolation should be considered to prevent damage of RPi. Read manual of your components.

When selecting transformer to match your valves voltage, check inrush current draw. PGV-100-GB has inrush 0.37A, and hold 0.21A. TRM-24 has 0.62A max current output.

Power coming from 230V Ac source on this schematic is already protected in home electrical installation. If you don't have any protections in your circuit, you should install them.

It is recommended to power RPi with good grade power adapter as bad one can cause micro shut downs.

To look at example connection made in real life, go to pictures folder.